Home Ice Advantage on Powerplay Performance in the NHL

Tim Campbell and Rafe Pasquini

Does Home Ice Advantage Affect Power Play Success?

Why this matters:

- Power plays are crucial scoring opportunities
  - Gives 1 team a man advantage
- Home ice advantage is widely discussed but rarely quantified
- Understanding factors can inform coaching decisions

Key Stat: Teams score on ~27% of power plays - what drives this variation?

Data Overview

- NHL play-by-play data (2021-2025)
- Sample Size:
  - 40,970 power play opportunities
  - 5,603 games analyzed
  - 32 teams across 4 seasons
  - Variables: Home/away, shot quality, rest days, travel distance

Expected Goals Model

- Probability a shot becomes a goal based on location
- Features used:
  - Distance from goal
  - Angle to goal
  - Shot location (slot vs. perimeter)
  - Summed total xG per powerplay
- Model performance:
  - AUC: 0.71 (good discrimination)
  - Brier Score: 0.062 (well-calibrated)

Regression and Predictions

Variable	Definition	Expected Effect
Home Ice	Home = 1, Away = 0	Positive
XG Per Shot	Shot Quality Metric	Positive
Rest Days	Days Since Last Game	Positive
Back-to-Back	B2B Games (1/0)	Negative
Travel Distance	Miles Traveled (per 100 miles)	Negative

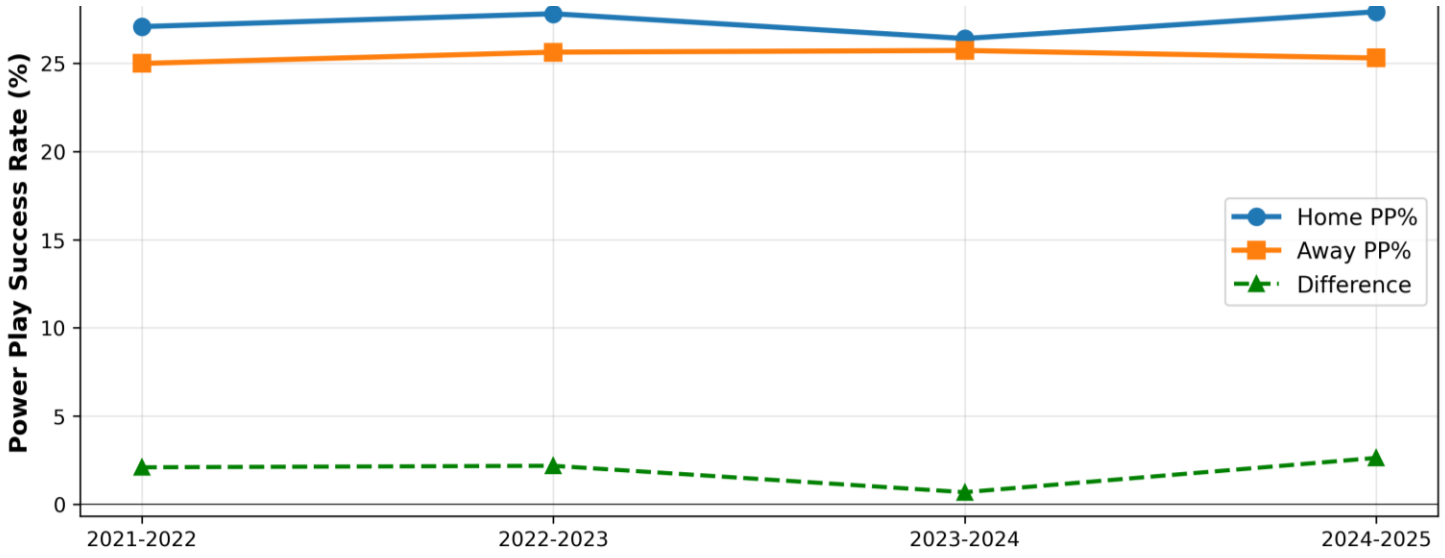
$$PP\% = \beta_0 + \beta_1(\text{Home}) + \beta_2(\text{xG/Shot}) + \beta_3(\text{Rest Days}) + \beta_4(\text{Back-to-Back}) + \beta_5(\text{Travel}) + \text{Team FE} + \text{Season FE} + \varepsilon$$

Main Regression Result

- Shot quality (xG): +5.54 per unit increase
- Home Games: +1.32 every home game
- Rest days: +0.017 per additional rest day
- Back-to-back penalty: -0.353
- Travel distance: Small positive effect
- Low R^2 of 0.008
- Large sample size (n = 40,789) validates model despite incredibly rest variable being minimal at the individual level

```
==============================================================================
MODEL 1: OLS REGRESSION - PP% ~ Home + xG + Rest Days + Travel
==============================================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:                 pp_pct   R-squared:                       0.008
Model:                            OLS   Adj. R-squared:                  0.008
Method:                 Least Squares   F-statistic:                     57.90
Date:                Sun, 16 Nov 2025   Prob (F-statistic):           3.02e-60
Time:                        14:07:17   Log-Likelihood:            -1.4504e+05
No. Observations:               40703   AIC:                         2.901e+05
Df Residuals:                   40697   BIC:                         2.901e+05
Df Model:                           5
Covariance Type:                  HC3
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept         25.7045      0.097    263.917      0.000      25.514      25.895
is_home            1.3185      0.087     15.081      0.000       1.147       1.490
xG_per_shot        5.5441      0.913      6.071      0.000       3.754       7.334
rest_days         -0.0167      0.008     -1.991      0.047      -0.033      -0.000
back_to_back      -0.3533      0.143     -2.475      0.013      -0.633      -0.073
travel_distance    0.0002   7.91e-05      3.150      0.002    9.41e-05       0.000
==============================================================================
Omnibus:                     8768.185   Durbin-Watson:                   0.166
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            81472.118
Skew:                           0.774   Prob(JB):                         0.00
Kurtosis:                       9.756   Cond. No.                     1.53e+04
==============================================================================
```

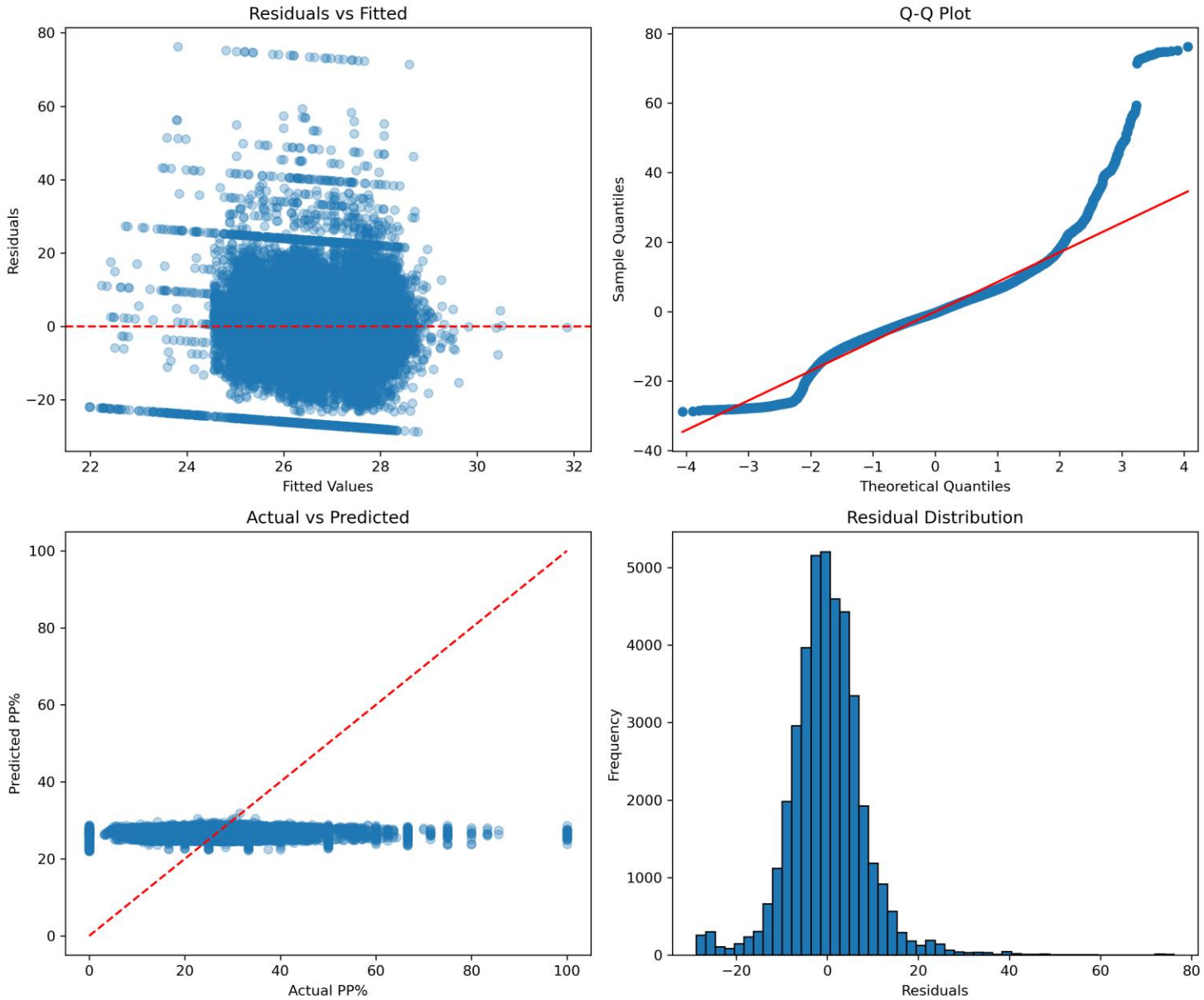
Controlling for Season



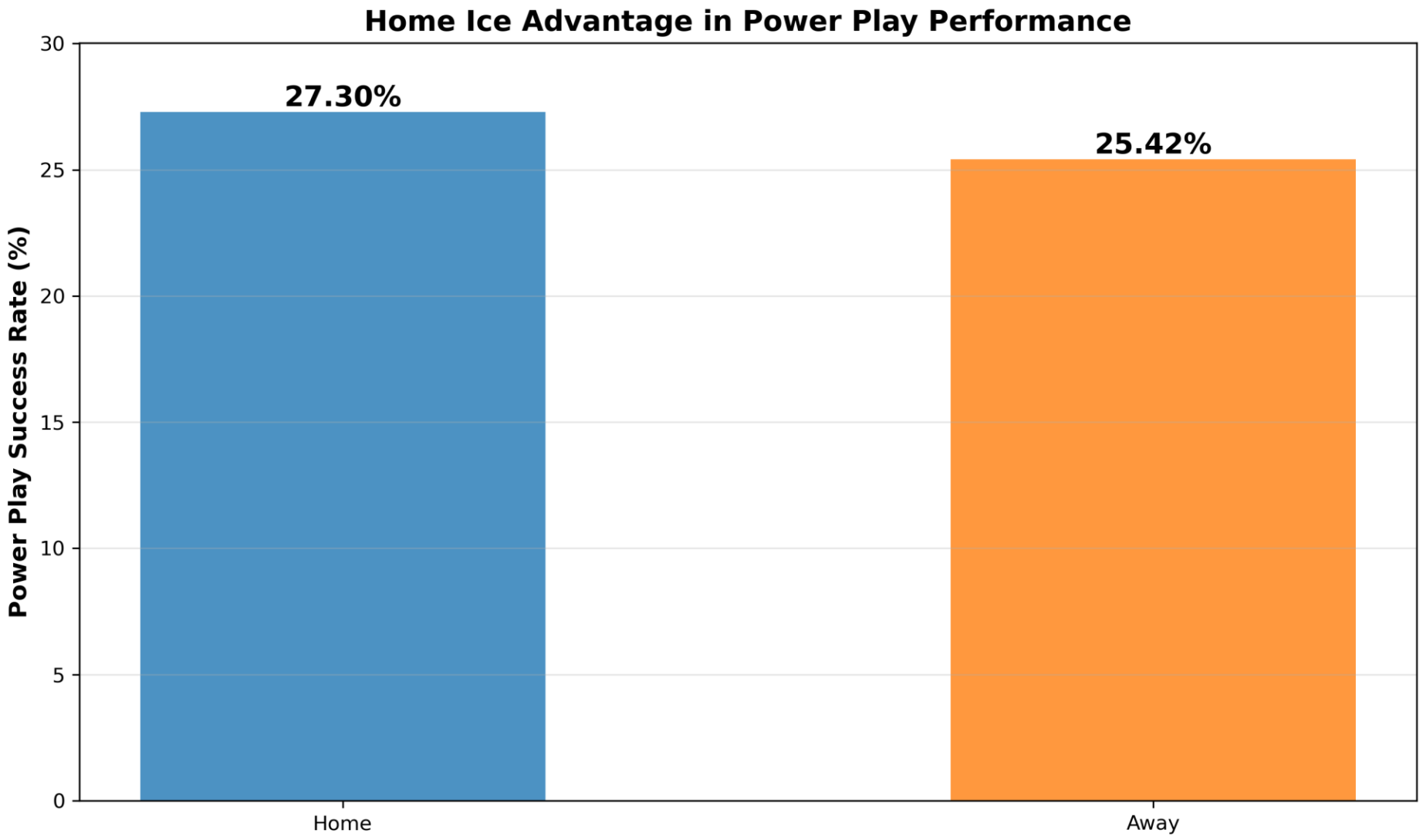
Season	Home PP%	Away PP%	Advantage
2021-2022	27.09%	24.99%	+2.09%
2022-2023	27.81%	25.64%	+2.18%
2023-2024	26.41%	25.73%	+0.68%
2024-2025	27.93%	25.30%	+2.62%

- Playing at home increases PP% by 1.31 percentage points
- Shot quality (xG): +7.66% per unit increase
  - Remains the strongest predictor even after controlling for season
- Back-to-back penalty: -0.037 per unit increase
  - Fatigue effect persists even when controlling for season trends
- Model fit improved: Adj. R^2 = 0.011 vs 0.008 in base model
  - Season effects explain additional variance in PP success

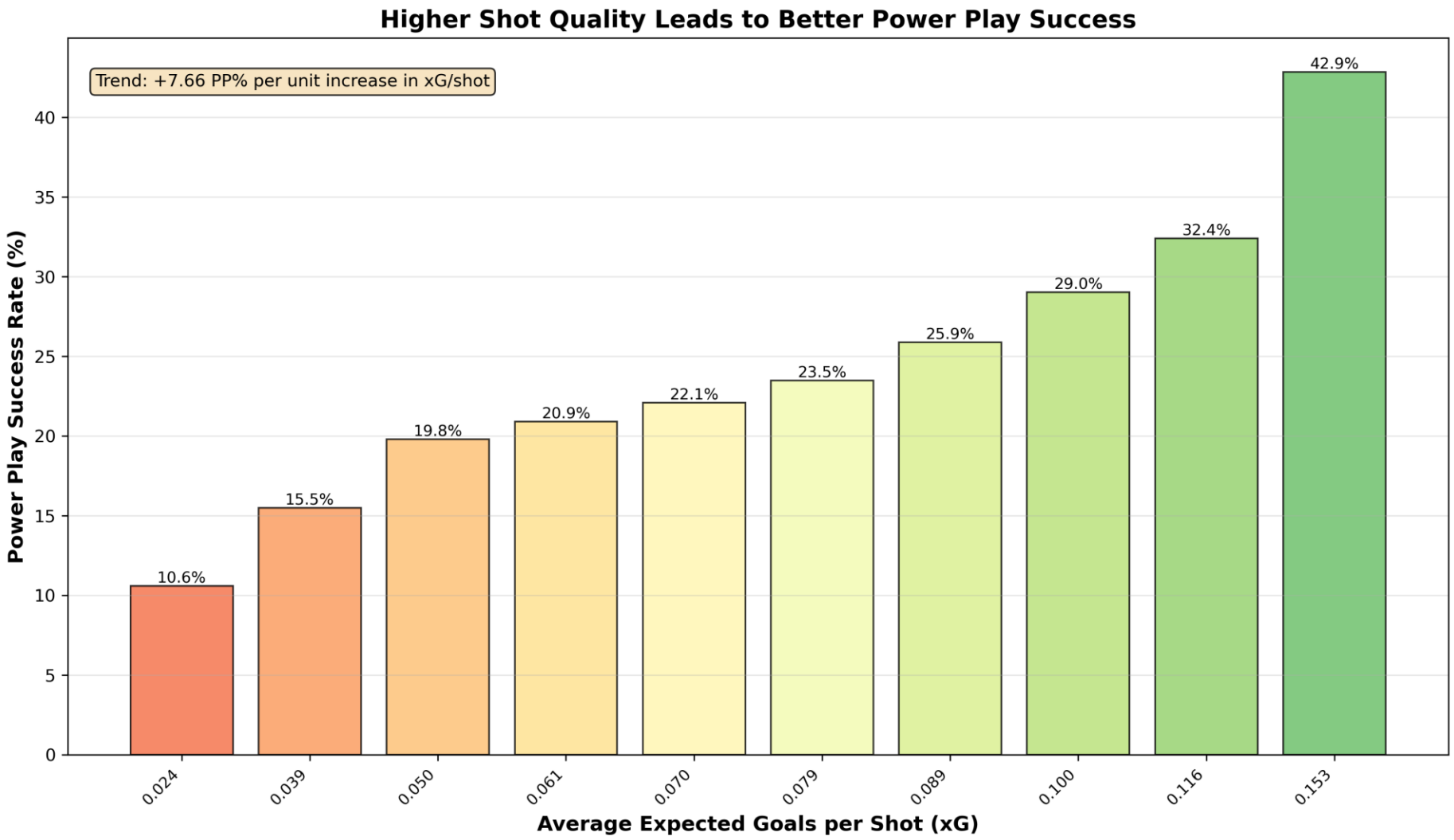
Model Diagnostics



Home vs Away

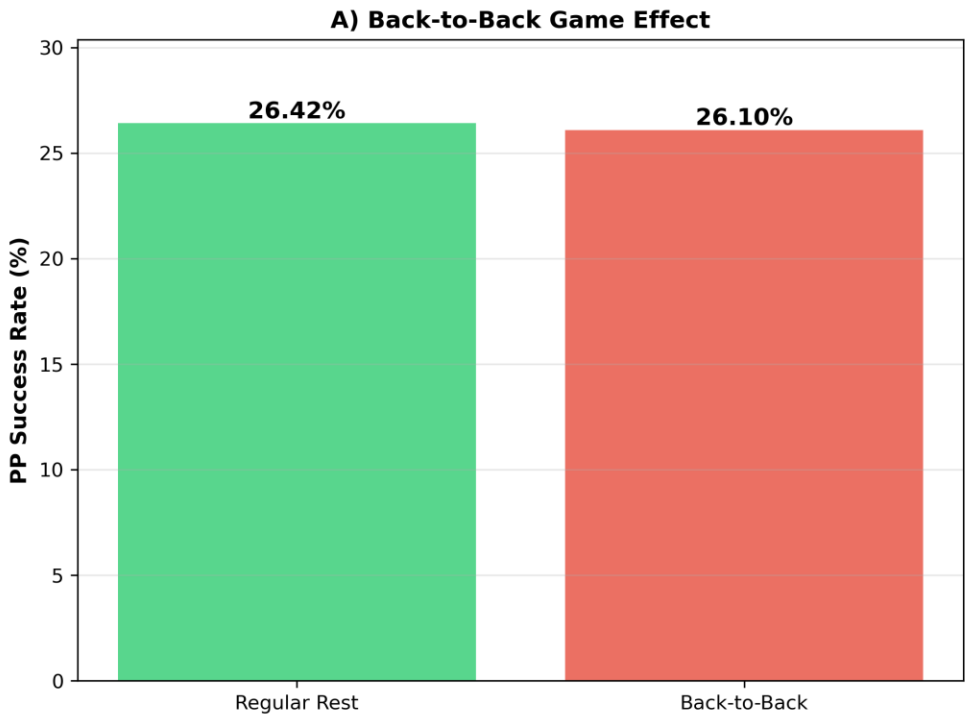

Home Ice Advantage in Power Play Performance
Power Play Success Rate (%)
30
25
20
15
10
5
0
27.30%
25.42%
Home
Away

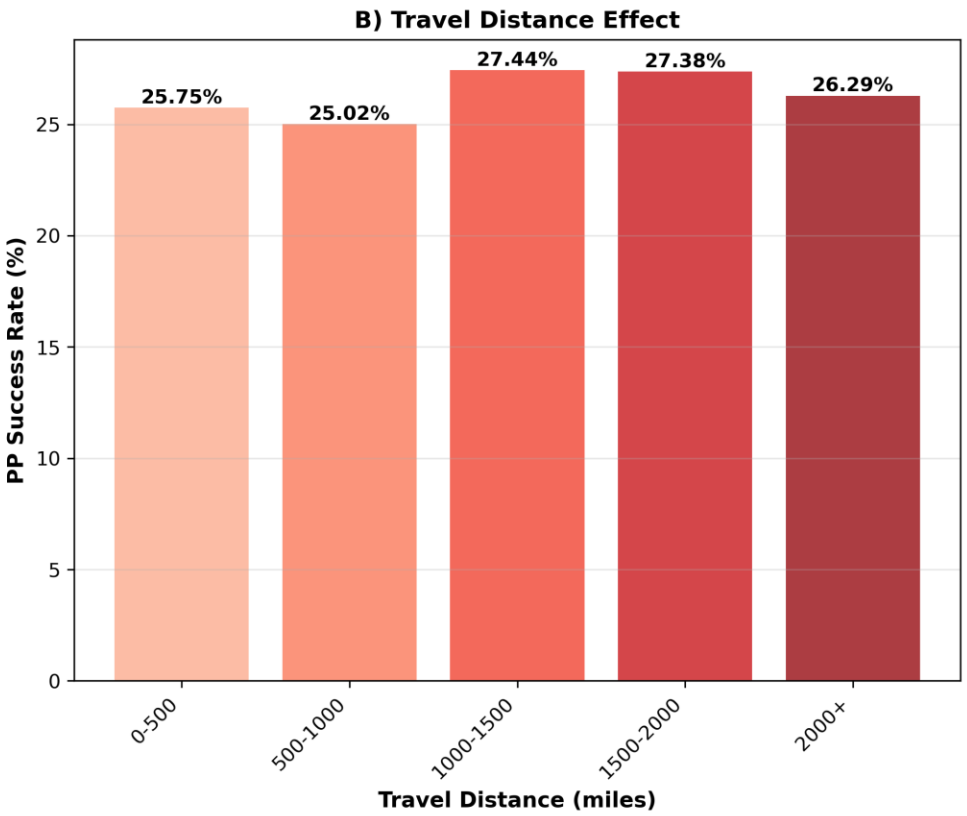
xG per Shot vs Powerplay



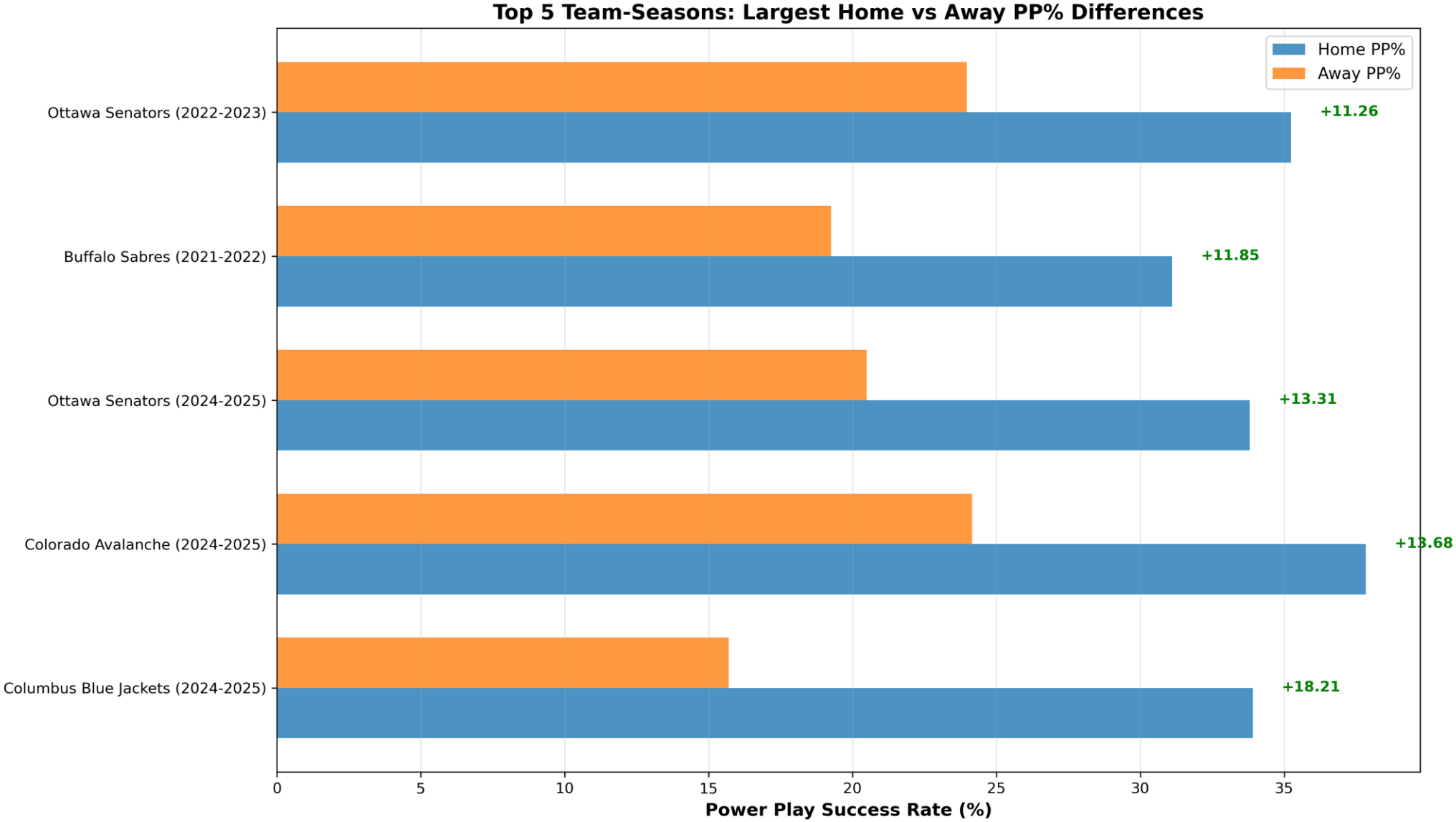
Fatigue and Travel Effects

- Back-to-back effect (-0.32%)
- Travel distance effect (-0.19% per 100 miles)
- Example: "1000-mile road trip costs ~2 PP percentage points"





Largest Difference in Our Data



Key Takeaways

- Home Ice Effect: +1.315%
  - Playing at home increases PP% by 1.31 percentage points
- Shot Quality Effect: +7.655
  - Each 0.01 increase in xG per shot increases PP% by 0.077 percentage points
- Rest Days Effect: -0.016%
  - Each additional day of rest changes PP% by -0.016 percentage points
- Back-to-Back Effect: -0.366%
  - Back-to-back games change PP% by -0.37 percentage points
- Travel Effect: +0.00025% per mile
  - Each 100 miles traveled changes PP% by 0.025 percentage points
- Model R-squared: 0.011